

Patch2Loc: Learning to Localize Patches for Unsupervised Brain Lesion Detection



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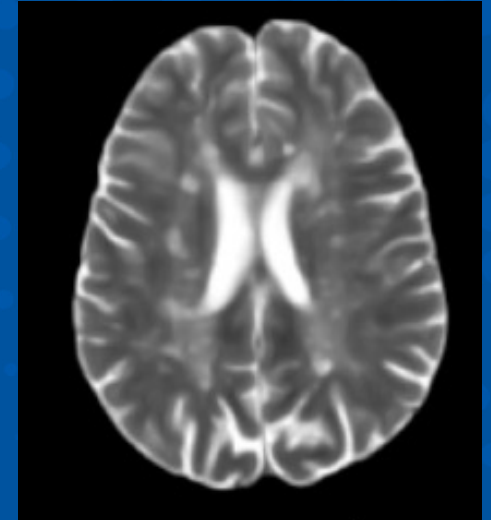
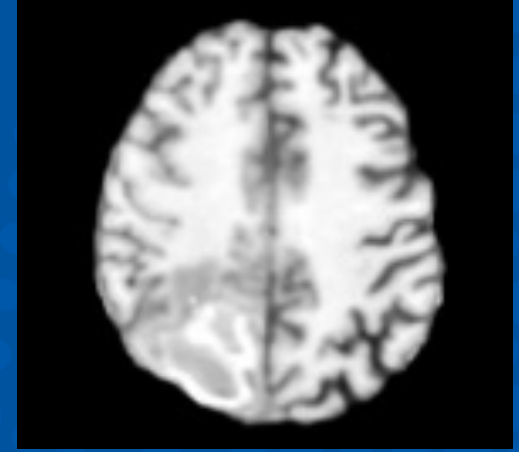
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Goal: Find abnormalities in neuroimages.

There is a lack of neuroradiologists and they are over worked.

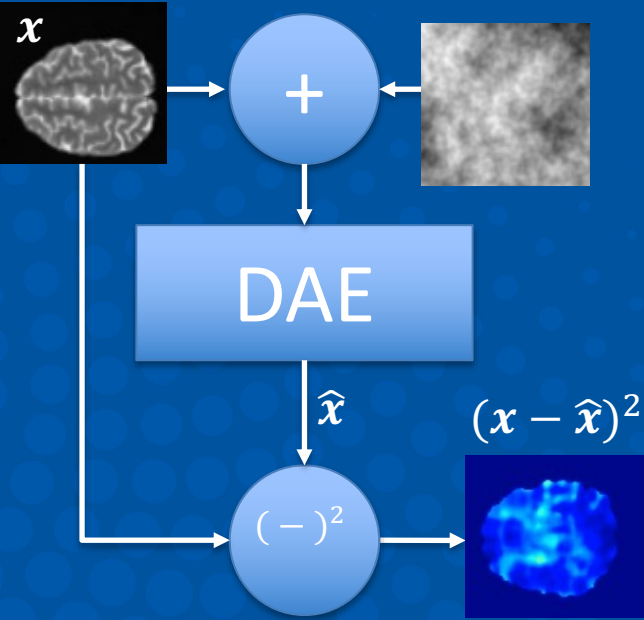
Could an unsupervised computer vision model **trained only with normal brains** help?

“All [normal brains] are alike; each [abnormal brain] is [abnormal] in its own way.” Tolstoy (1877)

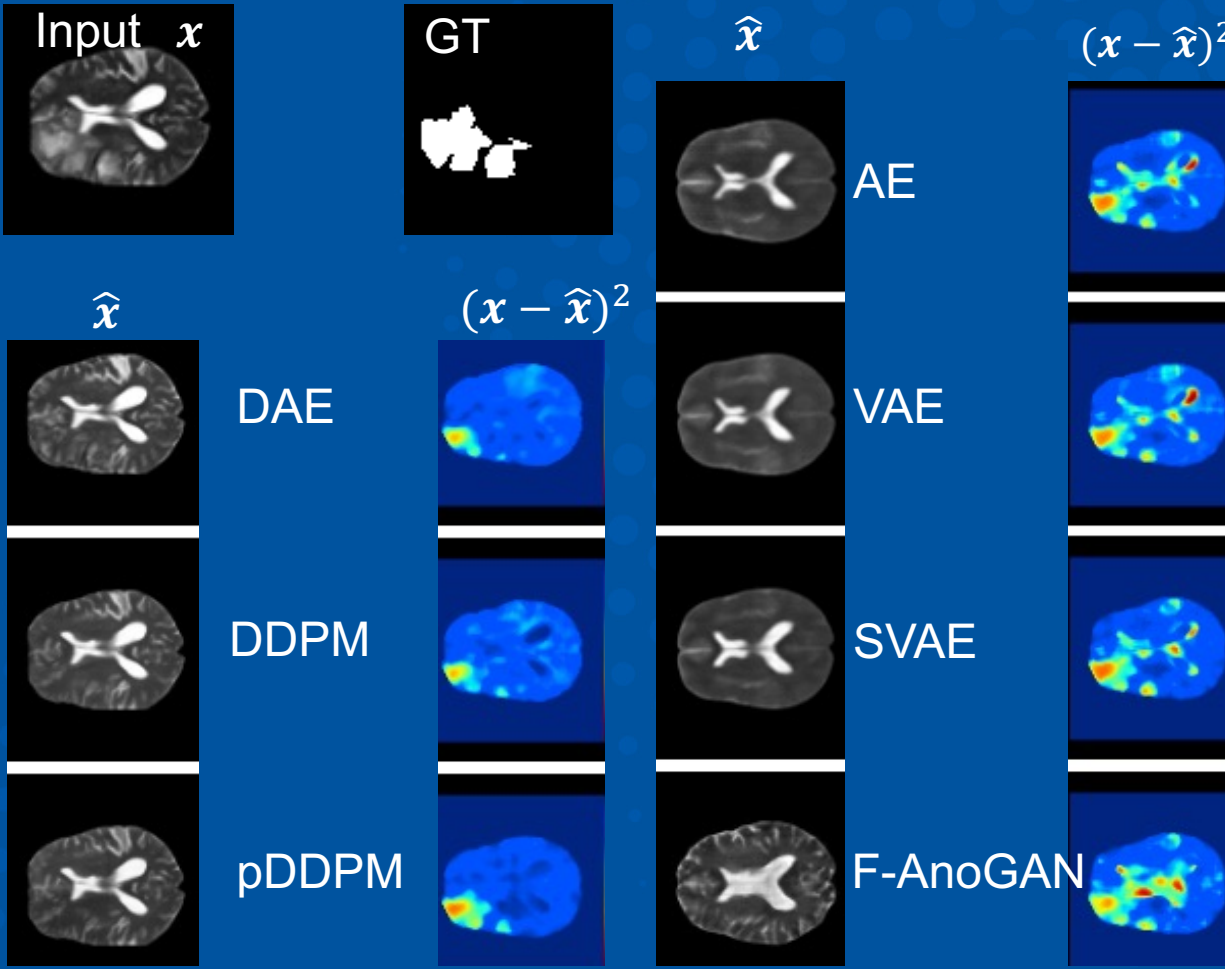


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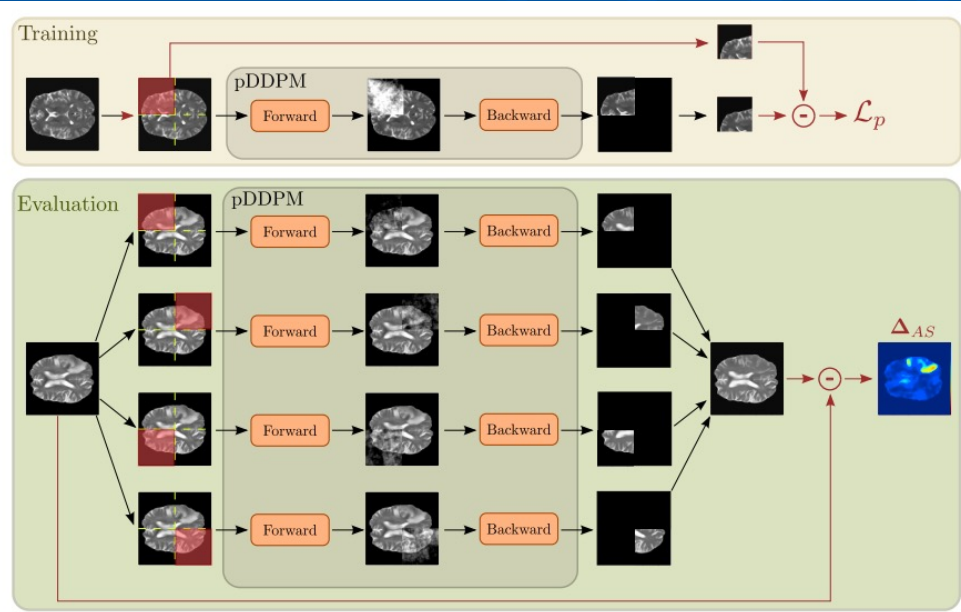
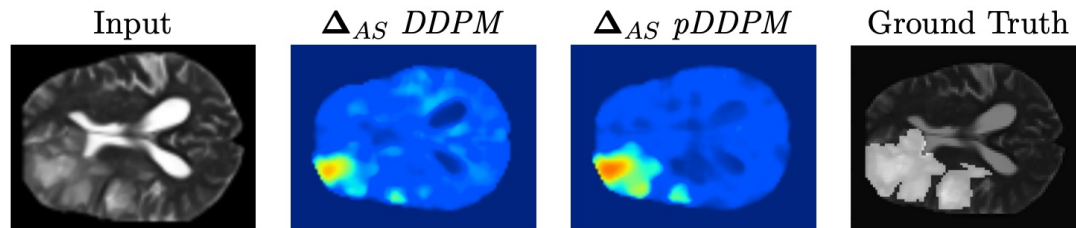
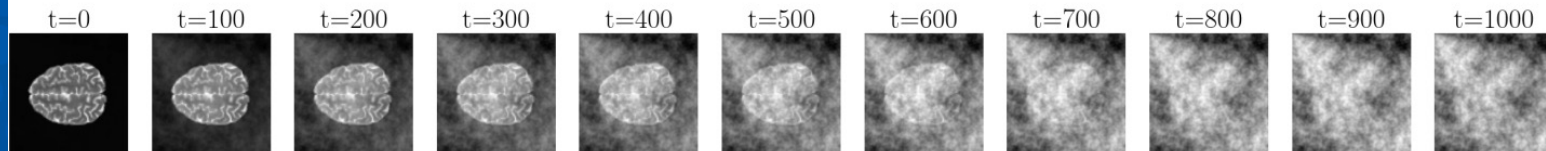
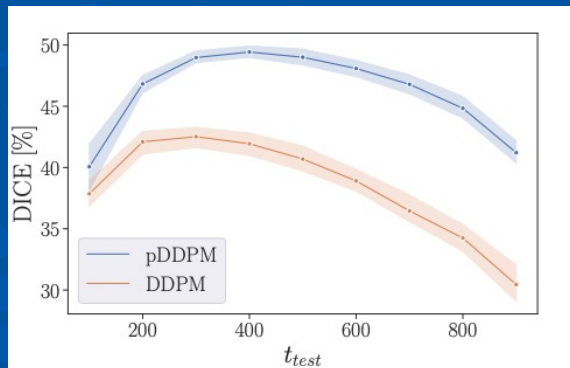
State-of-the-art and prior work uses global context of slice and (denoised) reconstruction



(Behrendt *et al.*, 2024)
“Patched Diffusion Models for
Unsupervised Anomaly...”



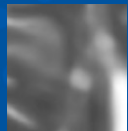
State-of-the-art adds localized noise, but is limited by noise paradox



(Behrendt *et al.*, 2024) “Patched Diffusion Models for Unsupervised Anomaly..”

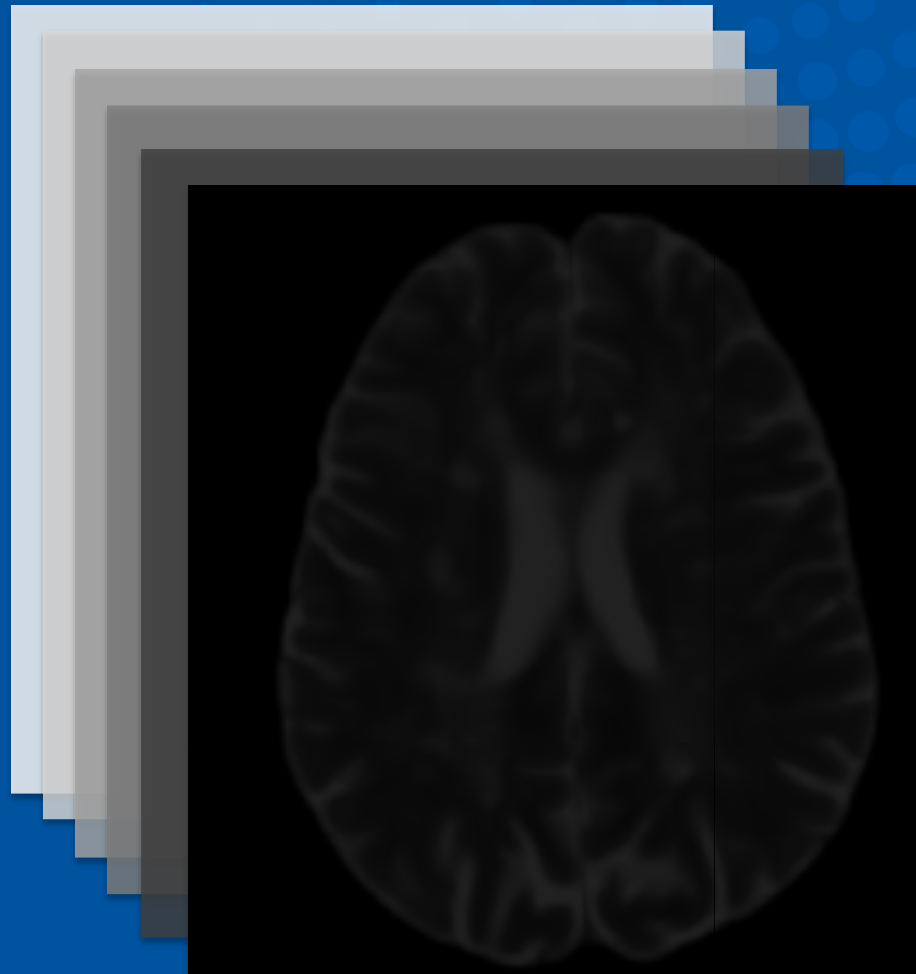
Our Insight:

Why not look locally?



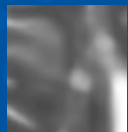
X is a patch

Patch2Loc asks, “Which location does this come from?”



Our Insights:

Why not look locally?



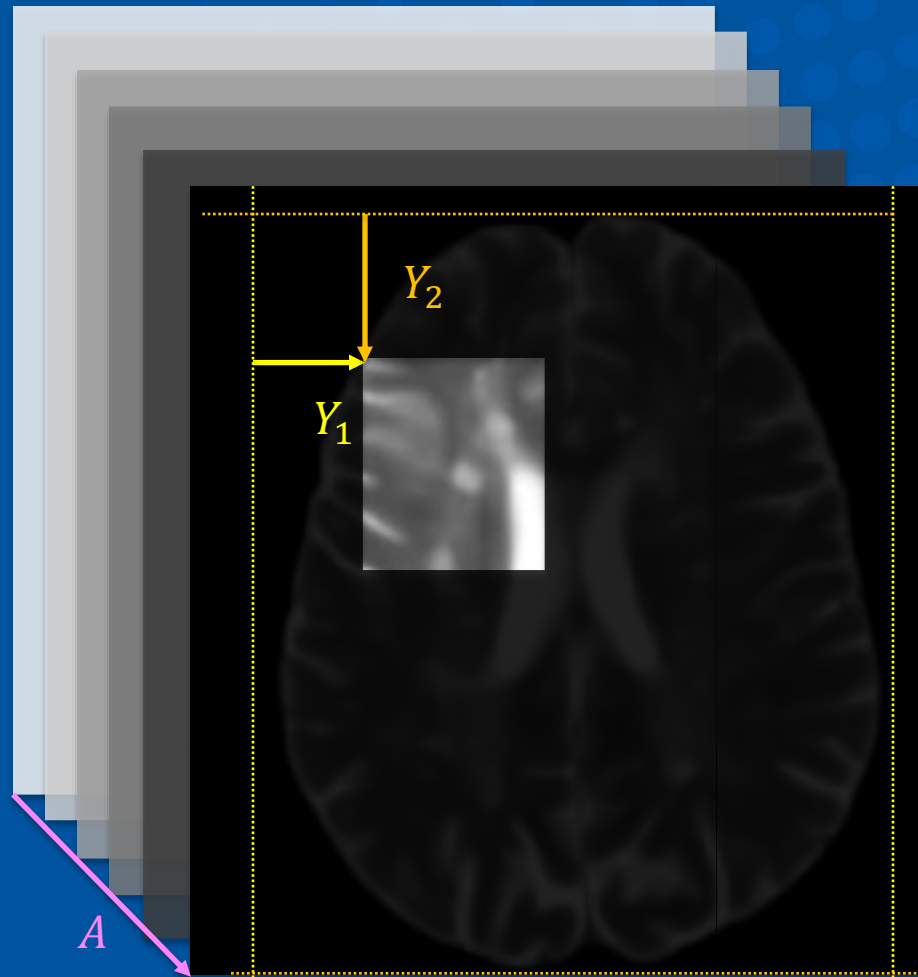
X is a patch

Patch2Loc asks, “Which location does this come from?”

$$P_{\theta}(Y|X, A)$$

Location $(Y, A) = (Y_1, Y_2, A) \in \mathbb{R}^3$

A axial coordinate

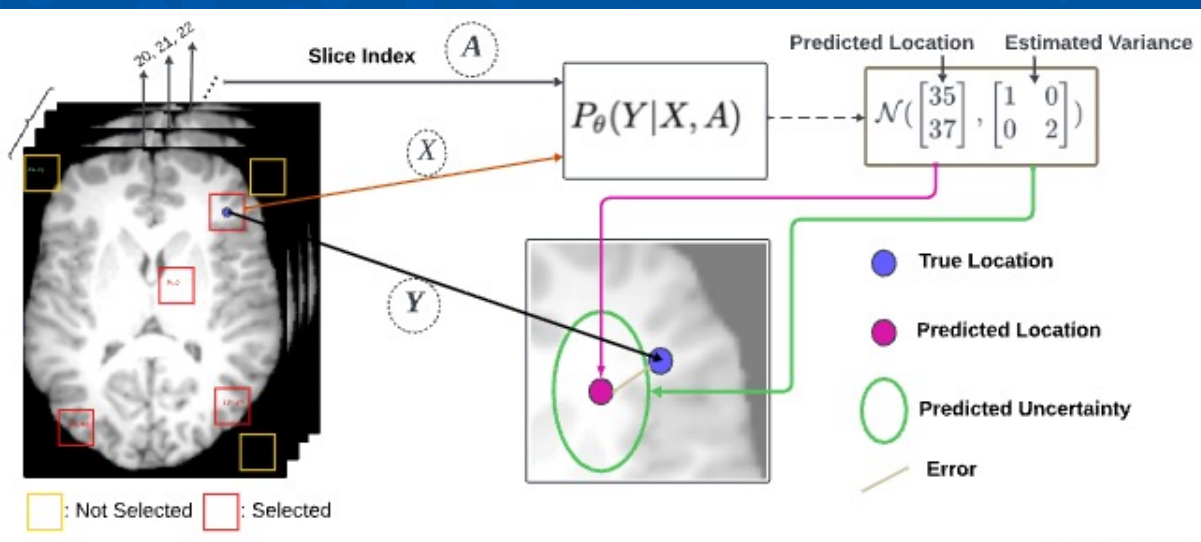


Patch2Loc Training via $\sim$ MLE of model relating Patch and Loc

$$P_{\theta}(Y|X, A) = \mathcal{N}(\mu^{\theta}(X, A), \Sigma^{\theta}(X, A)),$$
$$\Sigma^{\theta}(X, A) = \text{diag}([\exp(\varsigma_i^{\theta}(X, A))]_i)$$

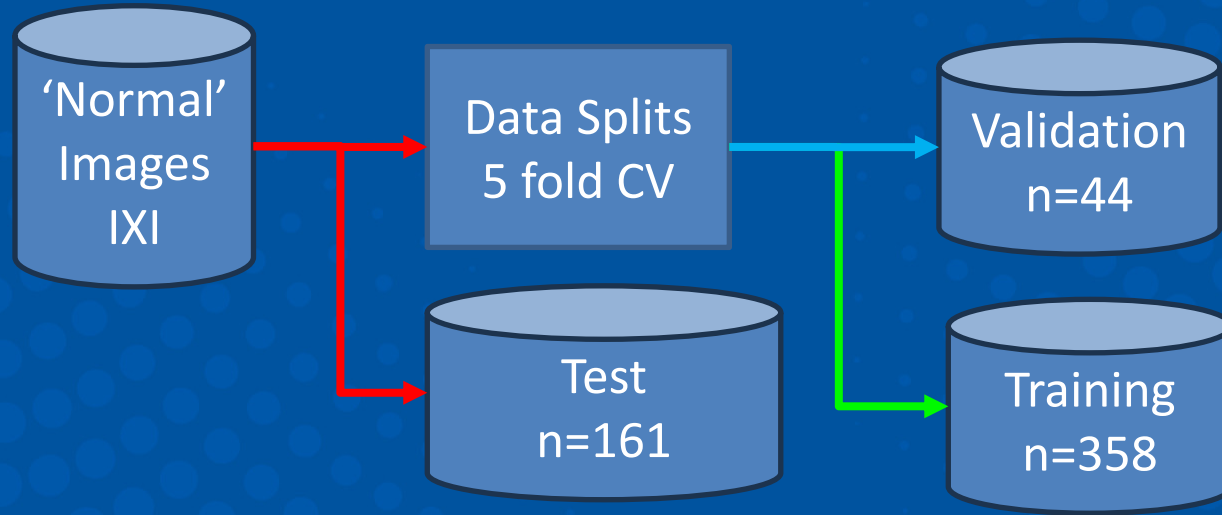
β -NLL [Seitzer et al., 2022] (for train and validation) where $[\cdot]$ is stop grad.

$$\min_{\theta} \mathbb{E}_{(X,Y,A)} \left[\sum_{i=1}^2 [\exp(\varsigma_i^{\theta}(X, A))]^{\beta} \left(\frac{|Y_i - \mu_i^{\theta}(X, A)|^2}{\exp(\varsigma_i^{\theta}(X, A))} + \varsigma_i^{\theta}(X, A) \right) \right]$$



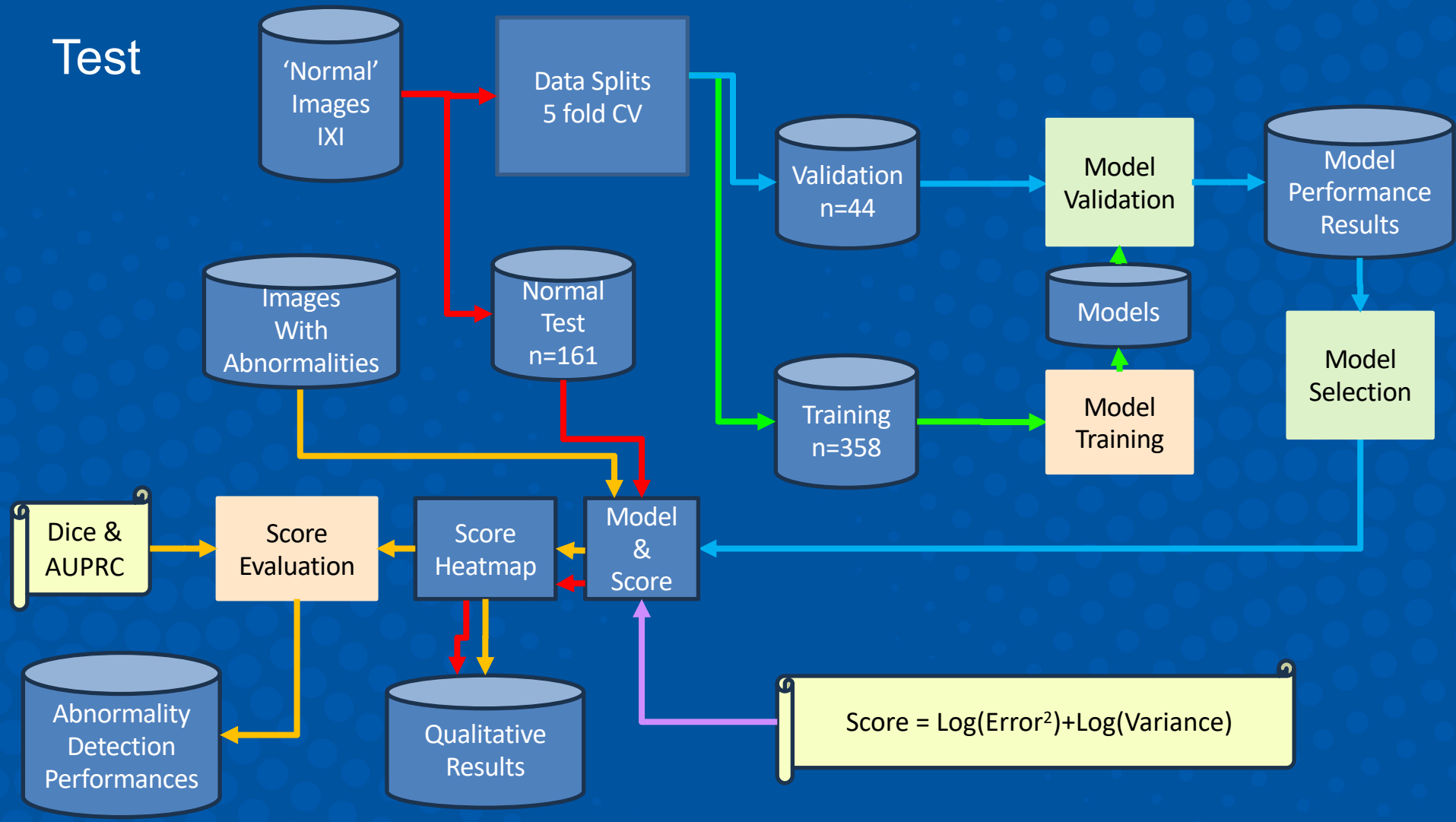
Patch2Loc Training

IXI dataset from Biomedical Image Analysis Group, Imperial College London



- Brain images are rigidly (affine) aligned and resampled.
- Histogram standardization (Nyúl et al., 2000) and normalization.
- Separate Patch2Loc model trained for T1 and T2

Test

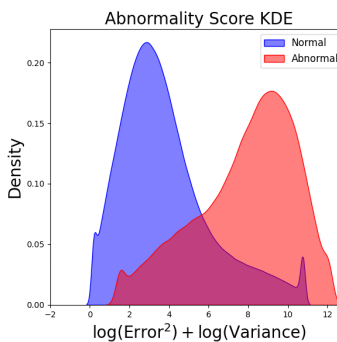
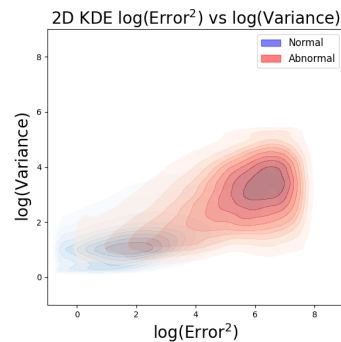
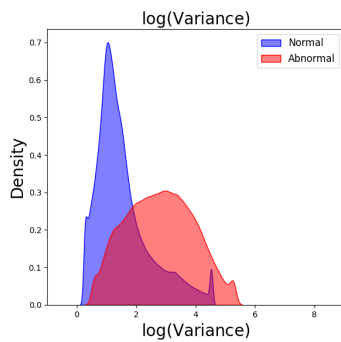
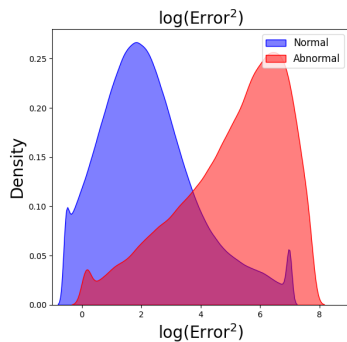
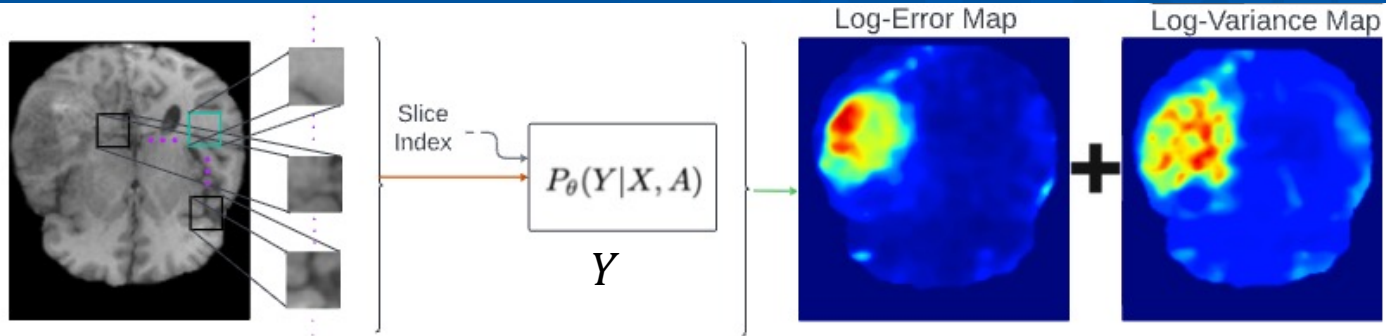
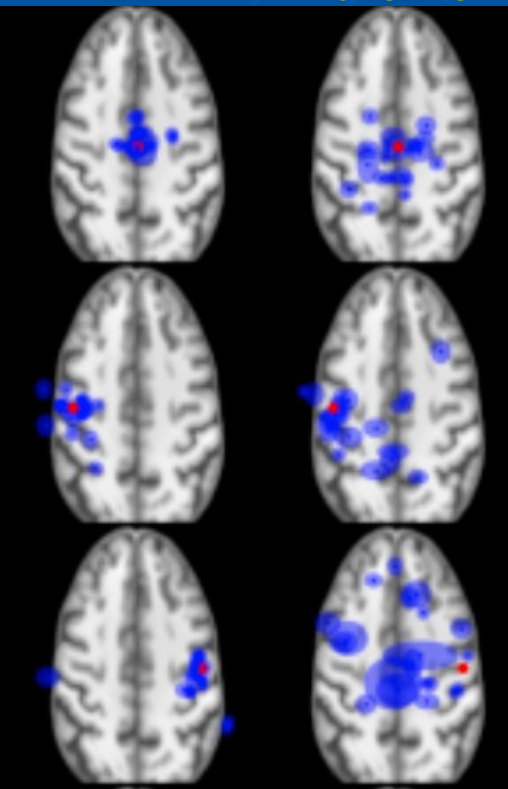


Inference: apply Patch2Loc convolutionally to make heatmaps

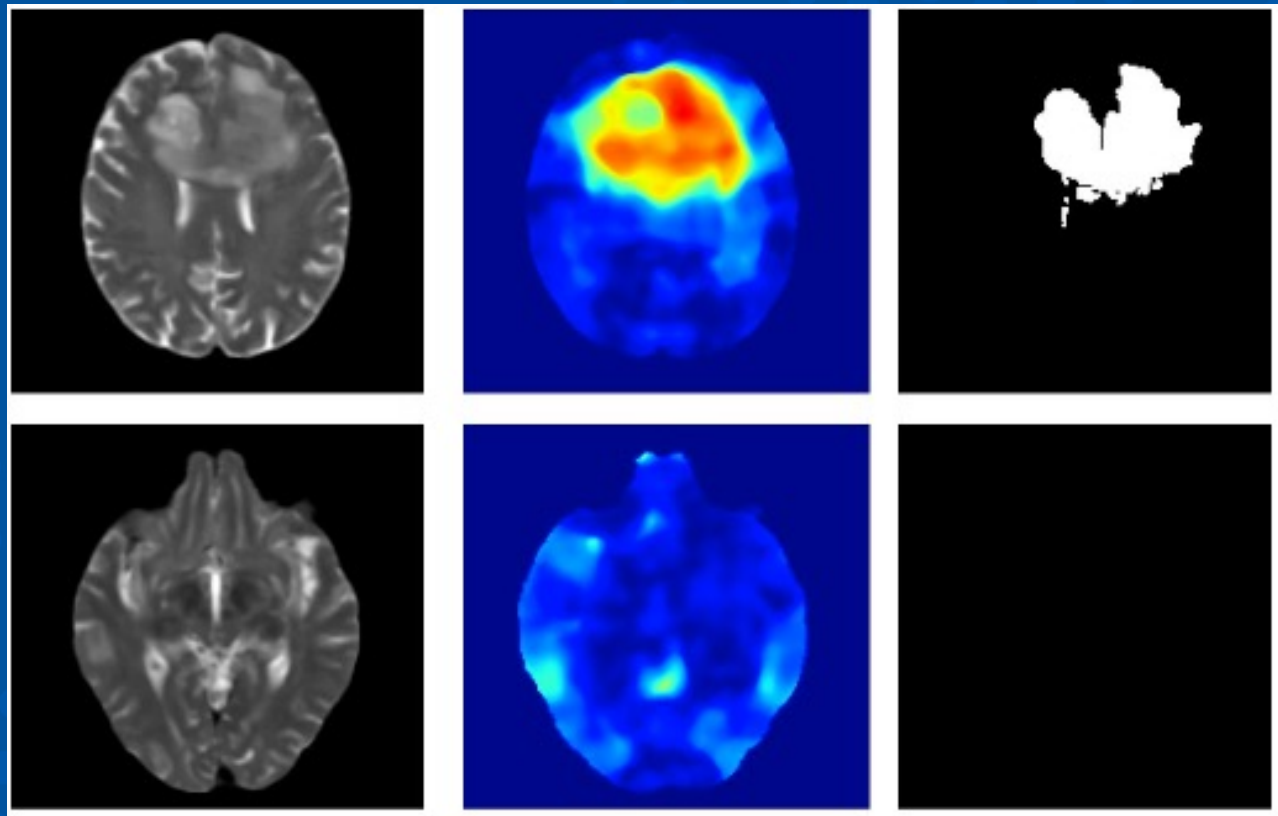
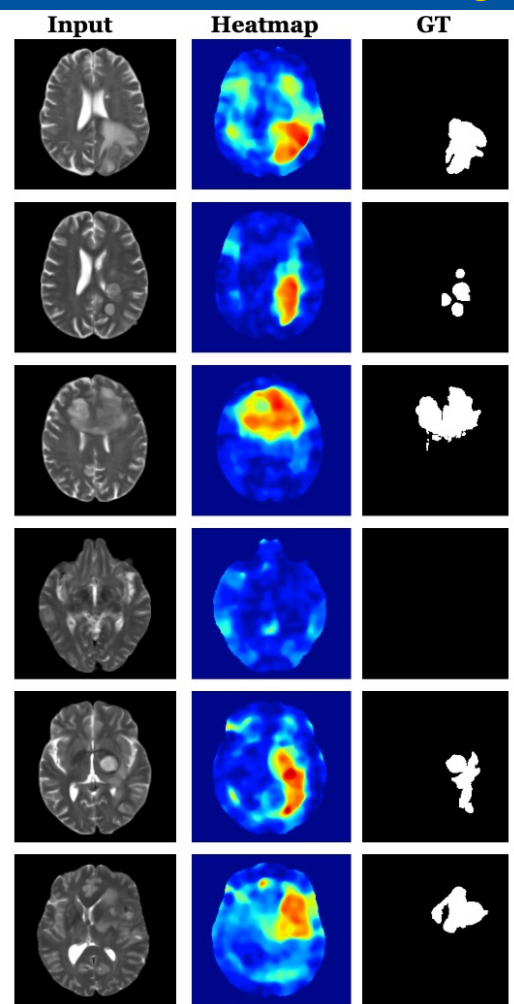
$$\text{Score}(X, Y, A) = \underbrace{\log(\|Y - \mu^\theta(X, A)\|_2^2 + \varepsilon)}_{\text{Log(Error}^2)} + \underbrace{\frac{1}{2}\varsigma_1^\theta(X, A) + \frac{1}{2}\varsigma_2^\theta(X, A)}_{\text{Log(Variance)}}$$

Normal

Abnormal



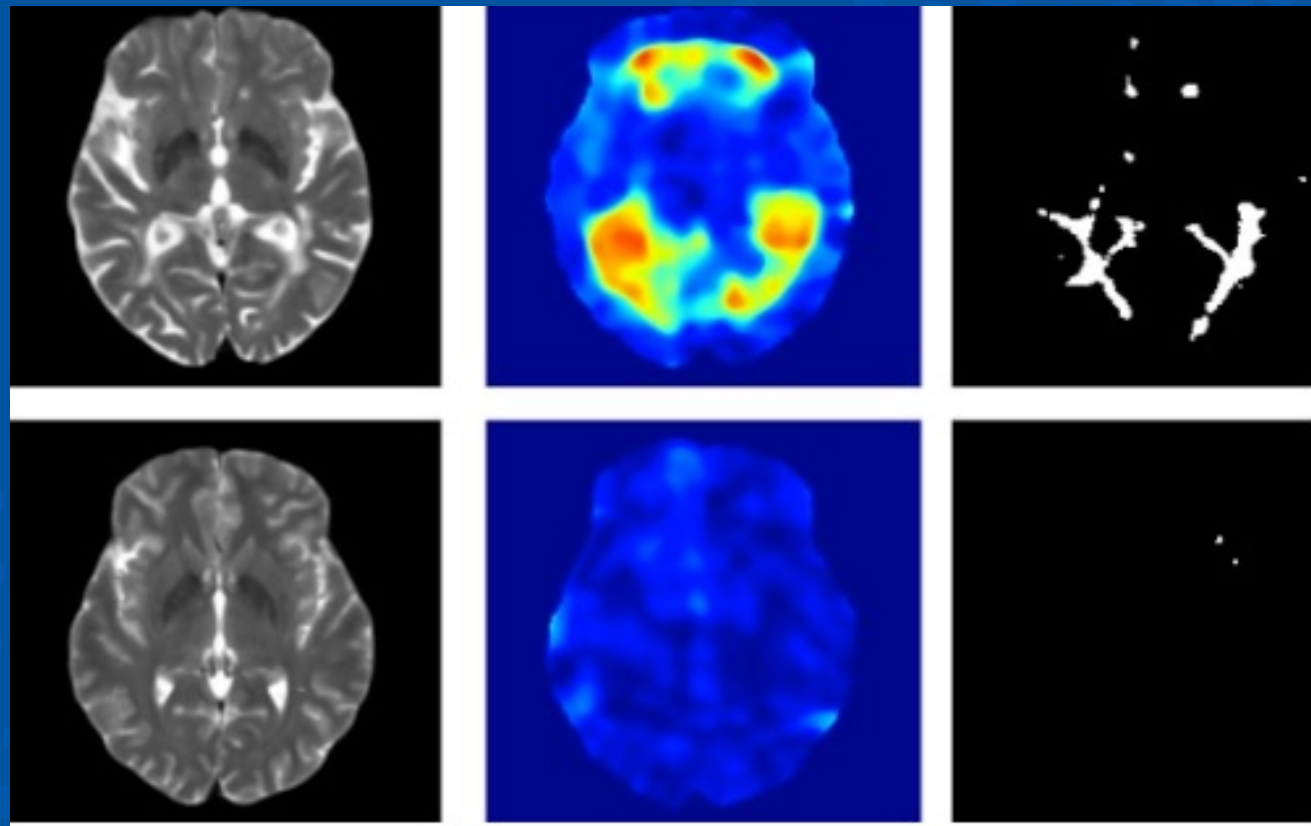
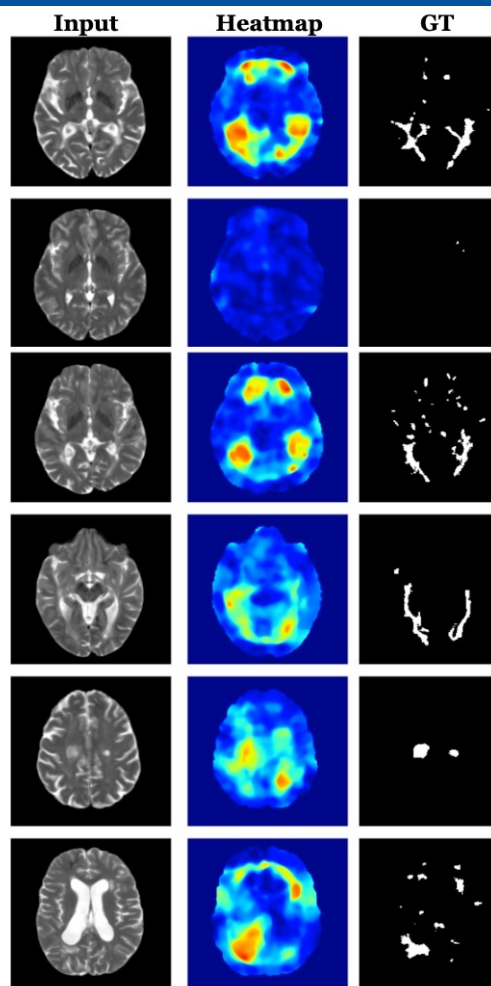
BraTS21, adult brain glioma, (Menze et al., 2015; Baid et al., 2021) n=1152



BraTS21, adult brain glioma, (Menze et al., 2015; Baid et al., 2021) n=1152

Model	BraTS21 (T2)	
	[DICE] [%]	AUPRC [%]
<i>Thresh</i> (Meissen et al., 2022a)	30.26	20.27
<i>VAE</i> (Baur et al., 2021)	33.12 $\pm$ 1.12	25.74 $\pm$ 1.37
<i>SVAE</i> (Behrendt et al., 2022)	36.43 $\pm$ 0.36	30.3 $\pm$ 0.45
<i>AE</i> (Baur et al., 2021)	36.04 $\pm$ 1.73	28.8 $\pm$ 1.72
<i>DAE</i> (Kascenas et al., 2022)	48.82 $\pm$ 3.68	49.38 $\pm$ 4.18
<i>RA</i> (Bercea et al., 2023b)	16.75 $\pm$ 0.51	9.98 $\pm$ 0.43
<i>PHANES</i> (Bercea et al., 2023c)	28.42 $\pm$ 0.91	21.29 $\pm$ 1.06
<i>FAE</i> (Meissen et al., 2022b)	44.59 $\pm$ 2.19	43.63 $\pm$ 0.47
<i>DDPM</i> (Wyatt et al., 2022)	50.27 $\pm$ 2.67	50.61 $\pm$ 2.92
<i>pDDPM</i> (Behrendt et al., 2024)	53.61 $\pm$ 0.51	55.08 $\pm$ 0.54
<i>cDDPM</i> (Behrendt et al., 2025)	56.30 $\pm$ 1.25	58.82 $\pm$ 1.56
<i>Patch2Loc</i> (Ours)	59.50 $\pm$ 1.45	55.40 $\pm$ 1.30

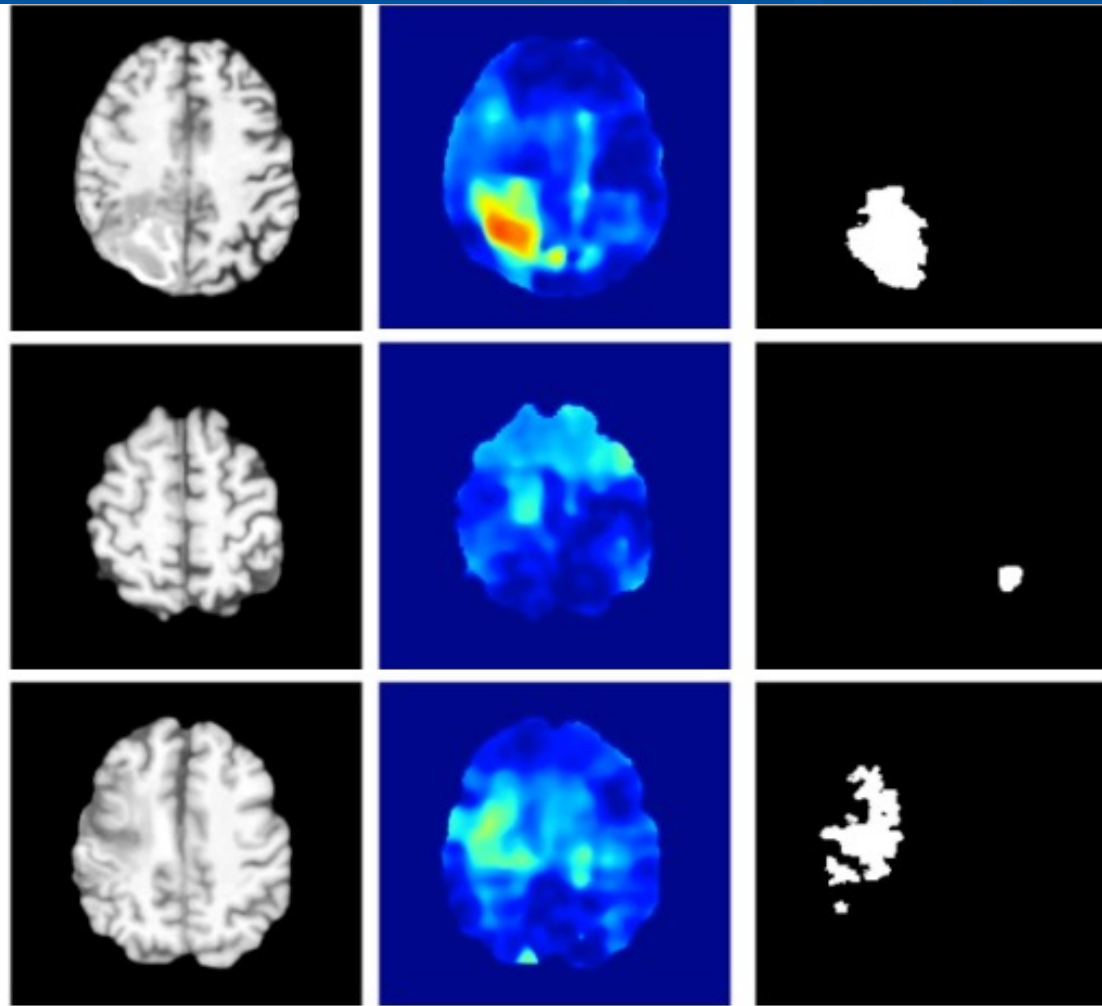
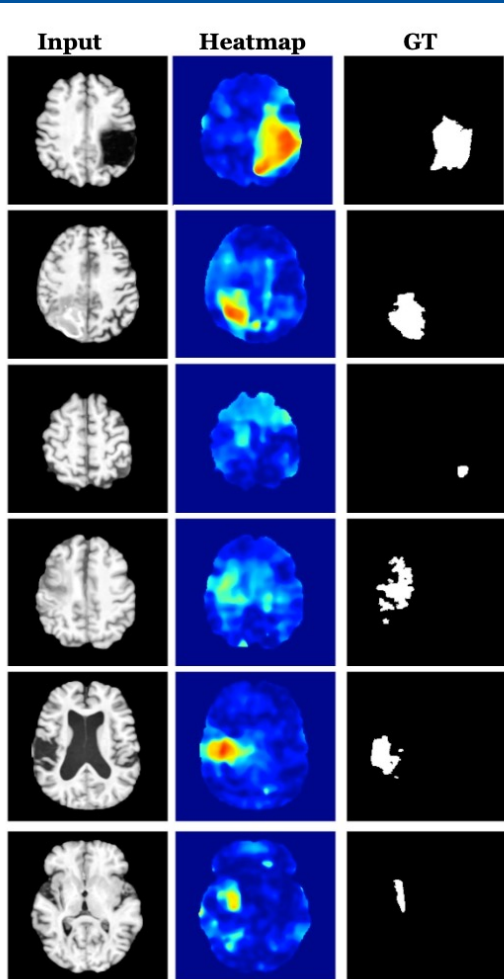
MSLUB, multiple sclerosis, (Lesjak et al., 2018) n=30



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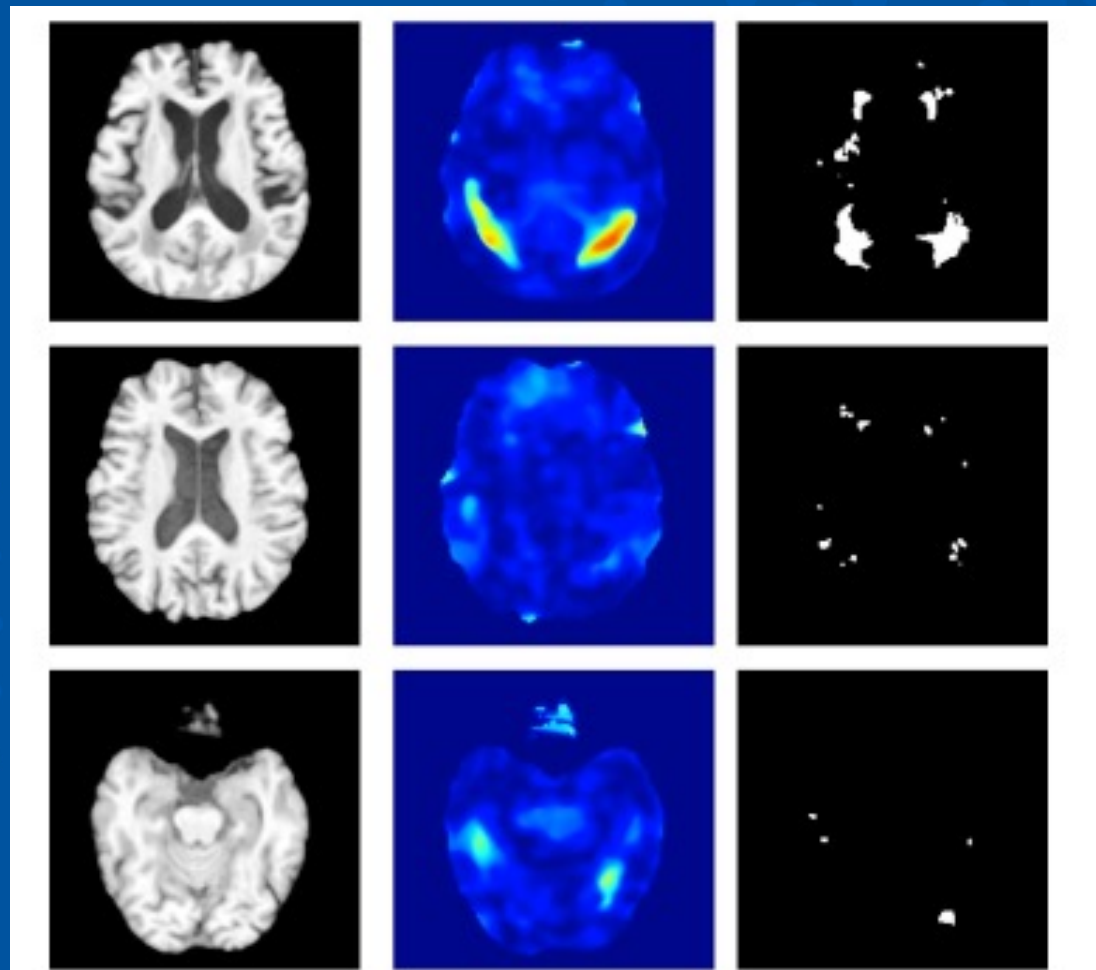
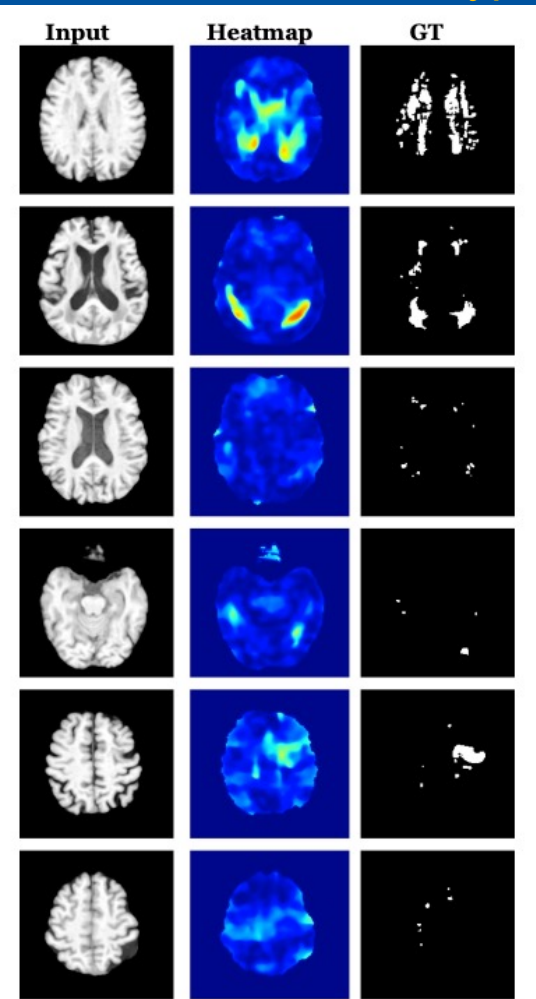
Model	MSLUB (T2)	
	[DICE] [%]	AUPRC [%]
<i>Thresh</i> (Meissen et al., 2022a)	7.65	4.23
<i>VAE</i> (Baur et al., 2021)	8.10 $\pm$ 0.18	4.48 $\pm$ 0.18
<i>SVAE</i> (Behrendt et al., 2022)	8.55 $\pm$ 0.11	4.8 $\pm$ 0.09
<i>AE</i> (Baur et al., 2021)	9.65 $\pm$ 0.97	5.71 $\pm$ 0.80
<i>DAE</i> (Kascenas et al., 2022)	7.57 $\pm$ 0.61	4.47 $\pm$ 0.69
<i>RA</i> (Bercea et al., 2023b)	3.96 $\pm$ 0.03	1.92 $\pm$ 0.04
<i>PHANES</i> (Bercea et al., 2023c)	6.11 $\pm$ 0.27	2.98 $\pm$ 0.07
<i>FAE</i> (Meissen et al., 2022b)	6.85 $\pm$ 0.65	3.85 $\pm$ 0.08
<i>DDPM</i> (Wyatt et al., 2022)	9.71 $\pm$ 1.29	6.27 $\pm$ 1.58
<i>pDDPM</i> (Behrendt et al., 2024)	12.83 $\pm$ 0.40	10.02 $\pm$ 0.36
cDDPM (Behrendt et al., 2025)	14.04 $\pm$ 1.16	10.97 $\pm$ 1.17
<i>Patch2Loc</i> (Ours)	14.30 $\pm$ 1.40	8.70 $\pm$ 1.20

ATLAS, Anatomical Tracings of Lesions After Stroke v2.0 (Liew et al., 2022) n=955



Model	ATLAS (T1)	
	[DICE] [%]	AUPRC [%]
<i>Thresh</i> (Meissen et al., 2022a)	4.66	1.71
<i>VAE</i> (Baur et al., 2021)	15.63 ± 0.73	11.44 ± 0.5
<i>SVAE</i> (Behrendt et al., 2022)	10.32 ± 0.53	6.84 ± 0.44
<i>AE</i> (Baur et al., 2021)	14.04 ± 0.6	10.16 ± 0.53
<i>DAE</i> (Kascenas et al., 2022)	15.95 ± 0.69	13.37 ± 0.62
<i>RA</i> (Bercea et al., 2023b)	12.21 ± 0.98	8.75 ± 0.93
<i>PHANES</i> (Bercea et al., 2023c)	17.62 ± 0.41	13.81 ± 0.48
<i>FAE</i> (Meissen et al., 2022b)	17.76 ± 0.16	13.91 ± 0.10
<i>DDPM</i> (Wyatt et al., 2022)	20.18 ± 0.58	17.77 ± 0.47
<i>pDDPM</i> (Behrendt et al., 2024)	19.92 ± 0.24	17.84 ± 0.10
<i>cDDPM</i> (Behrendt et al., 2025)	24.22 ± 1.10	22.22 ± 1.15
<i>Patch2Loc</i> (Ours)	25.50 ± 0.73	22.00 ± 1.70

WHM, white matter hyperintensity (Kuijf et al., 2019) n=60



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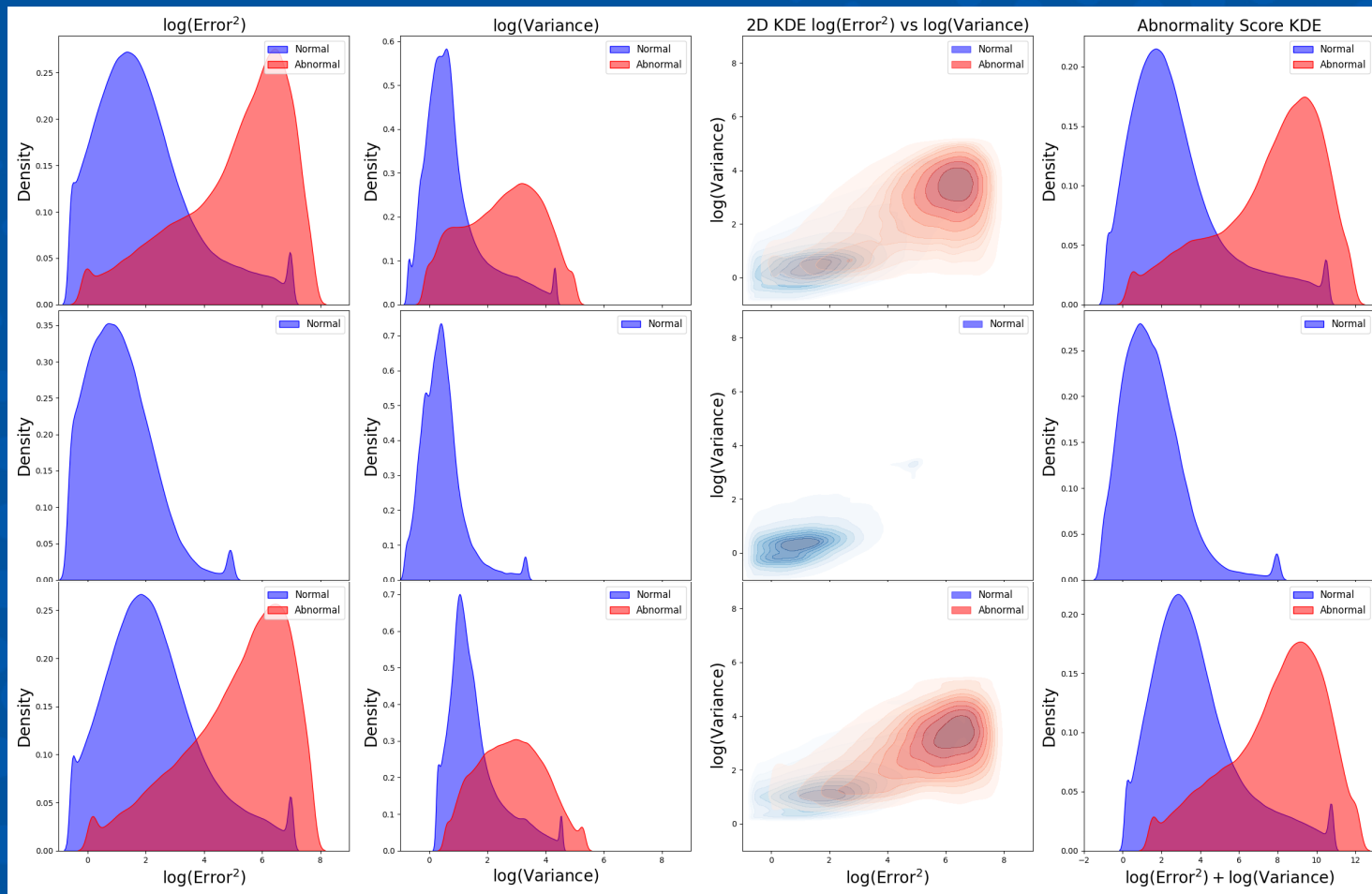
Model	WMH (T1)	
	[DICE] [%]	AUPRC [%]
<i>Thresh</i> (Meissen et al., 2022a)	10.32	4.72
<i>VAE</i> (Baur et al., 2021)	7.60 ± 0.28	3.86 ± 0.40
<i>SVAE</i> (Behrendt et al., 2022)	7.18 ± 0.07	2.97 ± 0.06
<i>AE</i> (Baur et al., 2021)	7.34 ± 0.08	3.43 ± 0.14
<i>DAE</i> (Kascenas et al., 2022)	12.02 ± 1.01	8.54 ± 1.02
<i>RA</i> (Bercea et al., 2023b)	6.04 ± 0.45	3.15 ± 0.31
<i>PHANES</i> (Bercea et al., 2023c)	7.55 ± 0.17	3.87 ± 0.13
<i>FAE</i> (Meissen et al., 2022b)	8.81 ± 0.38	4.77 ± 0.26
<i>DDPM</i> (Wyatt et al., 2022)	12.06 ± 0.97	8.89 ± 0.89
<i>pDDPM</i> (Behrendt et al., 2024)	10.13 ± 0.53	7.52 ± 0.56
<i>cDDPM</i> (Behrendt et al., 2025)	11.59 ± 0.93	9.26 ± 1.07
<i>Patch2Loc</i> (Ours)	15.70 ± 1.70	10.10 ± 1.70

Patch-wise Evaluation

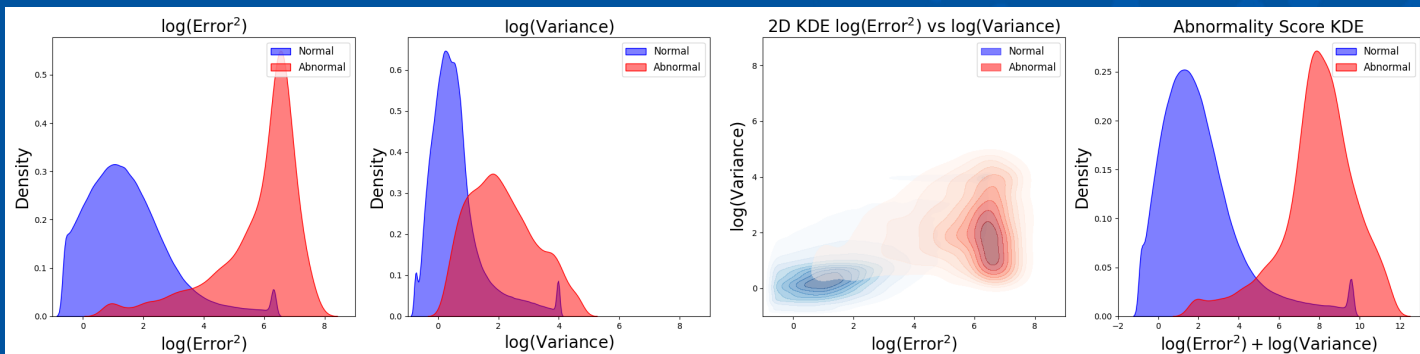
BraTS21 (T1)

IXI (T1) test

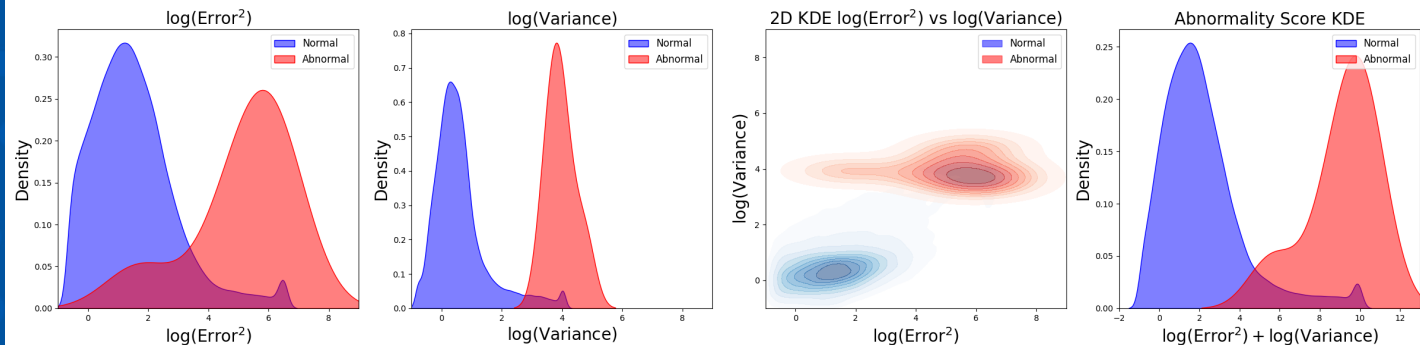
BraTS21 (T2)



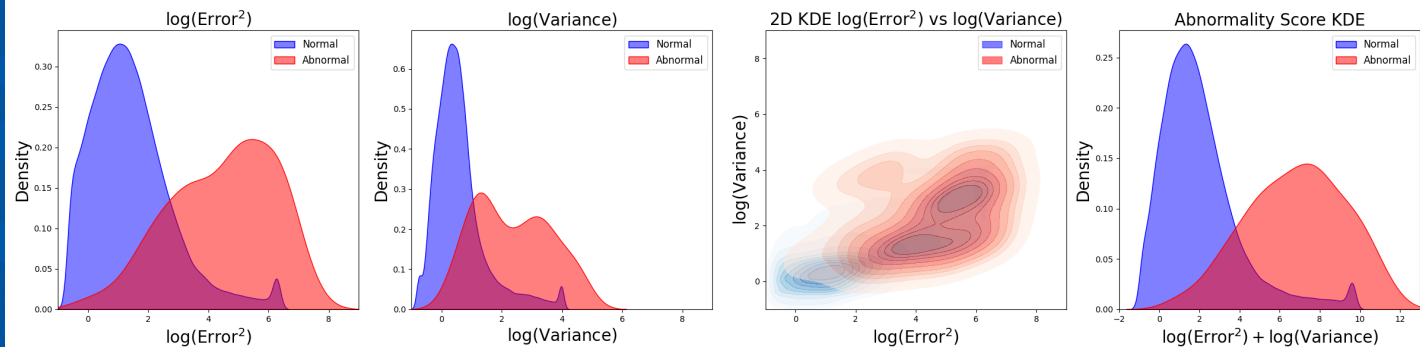
ATLAS



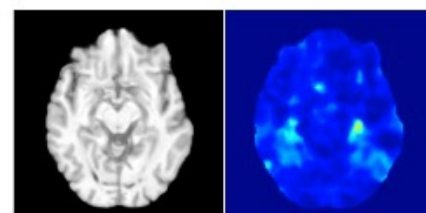
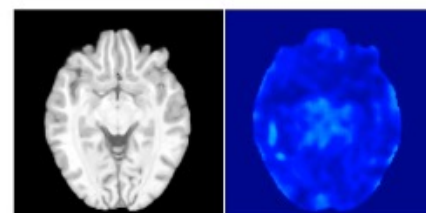
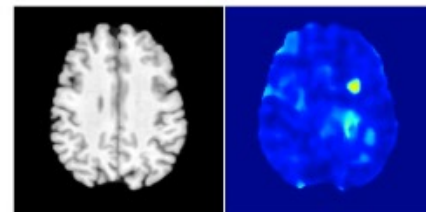
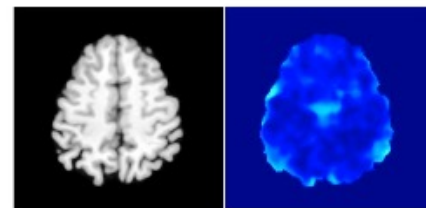
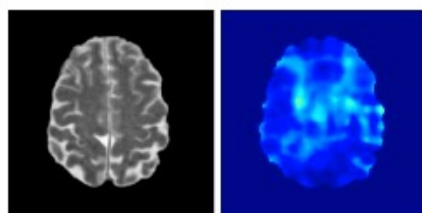
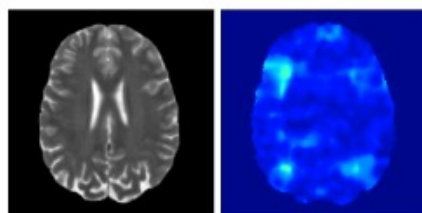
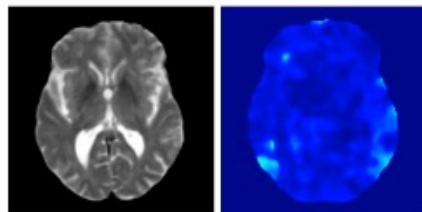
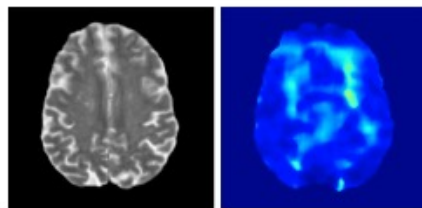
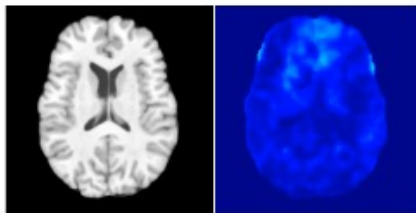
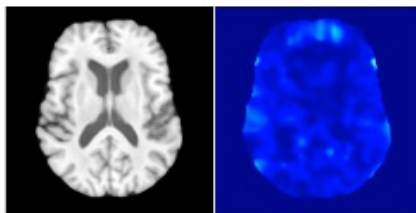
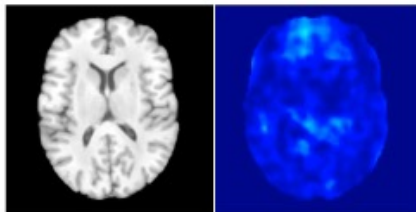
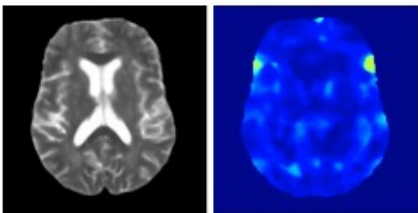
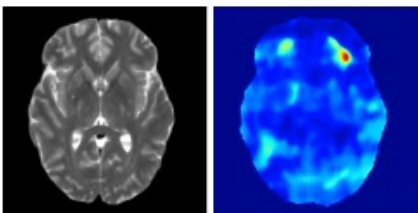
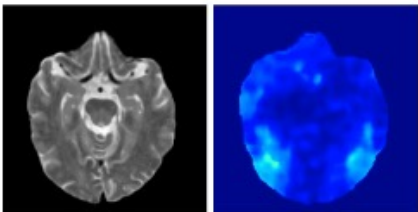
MSLUB



WMH



Healthy controls

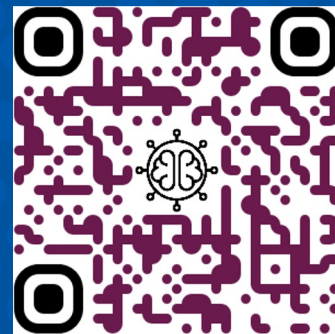


Future directions

Poster # 177

- Pediatric populations: sickle cell disease with silent infarcts, cortical malformations, epilepsy, TSC
- Multimodal fusion structural MRI (T1 and T2 and FLAIR)

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GitHub page



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